order28 RESULTS
PREREG sha256 033d322e64c07f3bd6ada371208a5cb624efa6b2303f78cfd2eb37b948013d3c (hashed before numerics).
Non-vacuity: d = relative distance from W = span{CO, Γ_γCO}, threshold 1e−3.
object	d	‖R‖ abs	rel	verdict
C1 corner(K)=CO		0.0		PASS
C2 R(CO) vs order25		54.406	0.0 err	PASS
subdivision K1+K2 = CO+¼vtx		2.2e−14		PASS (control)
subdivision J1+J2 = Γ_γCO+¼vtx		2.2e−14		PASS (control)
RG19 A2_lines LN	0.998	189.25	1.78	**FAIL**
RG20 A_K = Cor_{K1}−Cor_{K2}	0.580	224.90	0.147	**FAIL**
A_J = Cor_{J1}−Cor_{J2}	0.580	224.90	0.147	FAIL
K1 / K2 / J1 / J2	0.38 / 0.92 / 0.92 / 0.38	109.9 / 121.9 / 121.9 / 109.9	0.094 / 0.252 / 0.252 / 0.094	FAIL

Every test object passes non-vacuity: these are genuinely new directions.
RG19 explanation
‖R(A2)‖ = ‖R(LN+CO)‖ = 184.84 against ‖R(CO)‖ = 54.41. The lines do not cancel the corner defect; the full A2 is further from the law than its corner part. cos(R(LN), −R(CO)) = 0.224.
Prediction miss (disclosed): I predicted ‖R(LN)‖ ☑ ‖R(CO)‖ from a recalled order19 value. The order19 L0 value 54.4 was the corner/affine law, not the full A2. The prediction was wrong; the verdict FAIL stands.
RG20 explanation
The least-squares span of {R(A_K), R(A_J), R(LN)} captures 16.6% of ‖R(CO)‖². That is no explanation.
Post-hoc observation: J1 = γ⁻¹K2 and J2 = γ⁻¹K1 (γ⁻¹(1,1) = (20,−3)). So the mediant subdivision of γ⁻¹K is the γ⁻¹-transport of the K subdivision. The J objects are Γ_γ-images of the K objects (matching norms to 1e−13) and add nothing independent; this follows from the v8 transport proposition.
Category
A: nothing new (subdivision and transport identities were already in v8). B: all numbers above. v11: no. Ξ: not identified. E1–E3 OPEN.
Blind/consistency: RG20 blind (predicted FAIL, correct). RG19 prediction relied on a misremembered order19 value (wrong).

order28 NOTES
- Non-automorph cones within one γ-period are related by γ-transport (J = γ⁻¹K_i). Within the Farey tree only the K-subdivision family K1, K2, (K11, K12, …) is independent mod ⟨γ,h⟩.
- A2_lines fails the fixed-M weight-1 law badly (rel 1.78). The lines likely obey a different law (completion/shadow as for A1); "near-law" is a corner-only phenomenon.
- Next (order29, to preregister):
  (a) Test the A2_lines against the A1-type completed law (lines + completion), as in order17, rather than the fixed-M law.
  (b) Use a deeper Farey family K_{p/q} and look for a convergent combination Σ c_i R(Cor_{K_i}) cancelling R(CO), with c_i fixed by a geometric rule in advance (not fitted).
- DISCOVER_GRADING_PATH and MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim. nice, OMP_NUM_THREADS=1, 3 GB, ~55 s; no push or sends.